

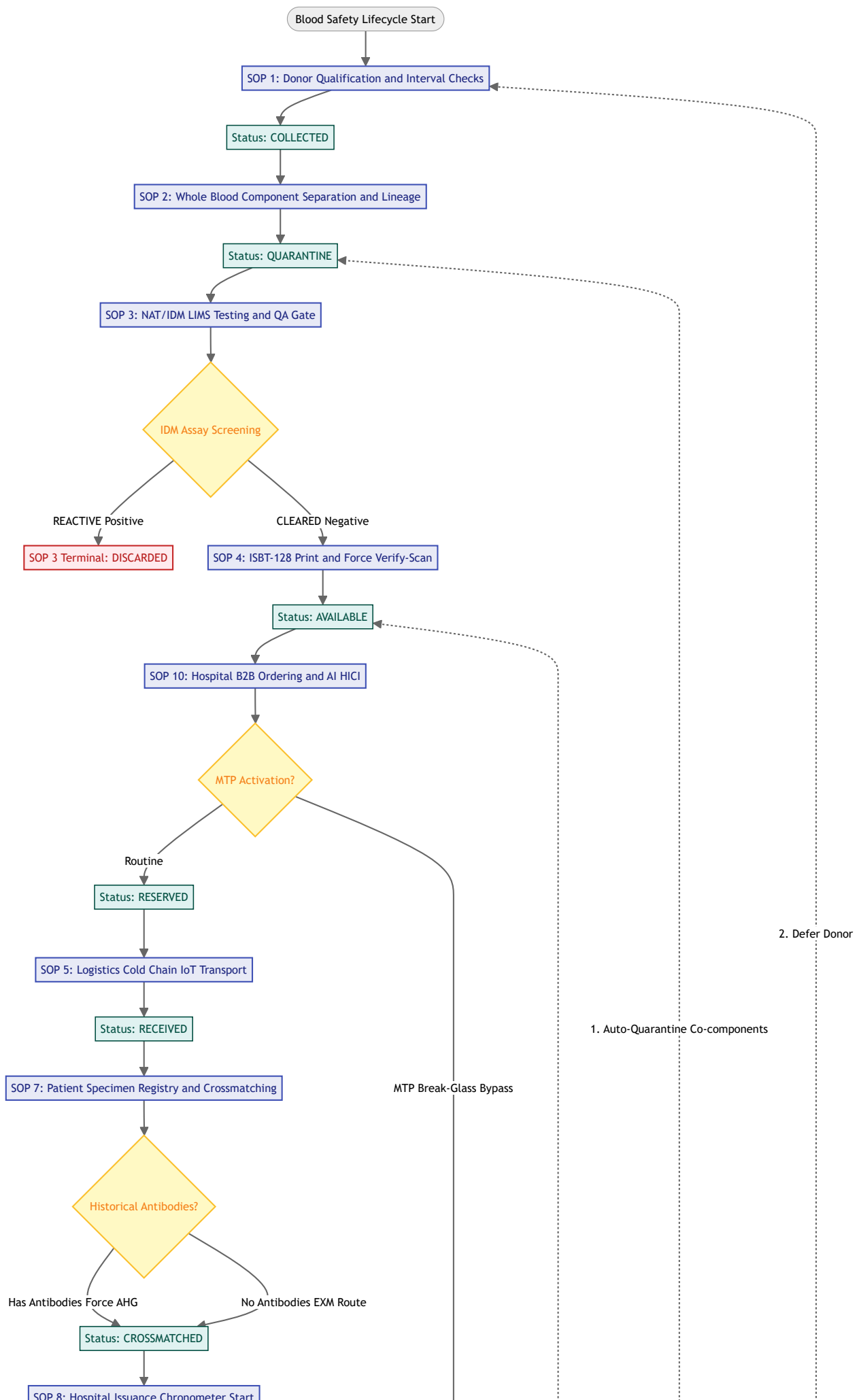
VN-BECS-V1 ALIGNMENT WHITEPAPER

SUBSYSTEM CROSS-CONNECTION & 10 SOP
INTEGRATION

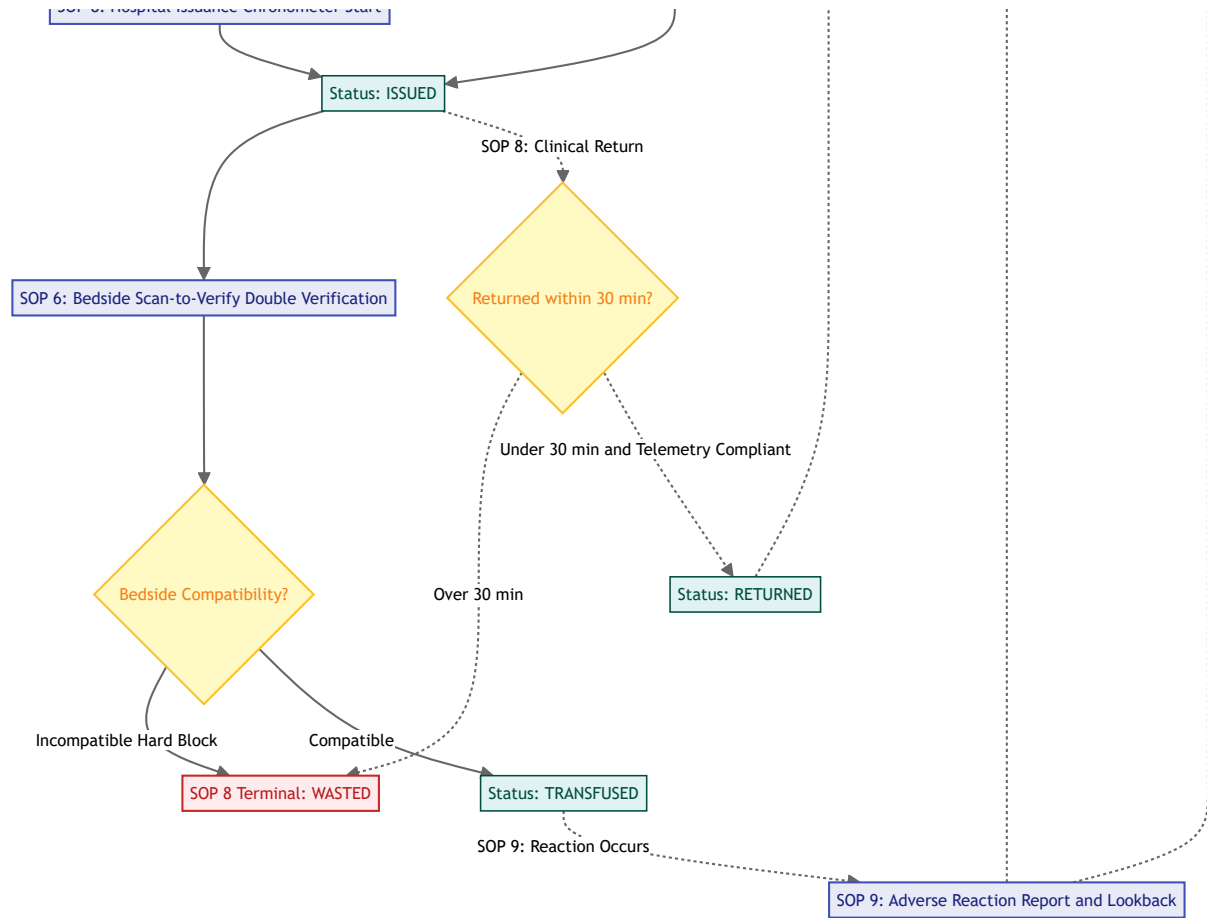
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CIRCULAR 26

I. SUBSYSTEM INTER-CONNECTIONS BLUEPRINT

II. 10 CORE SOPS OPERATIONAL LIFECYCLE DIAGRAM



SOP 5: Hospital Issuance Synchronization Start



III. SUBSYSTEM CORRELATION MATRIX

SUBSYSTEM	SOPS	SUBSYSTEM INTER-CONNECTION & SAFETY INTEGRITY
M1: Donor & LIMS	SOP 1, 3	Performs donor identity via CCCD card scan. Standardizes Circular 26 age/weight checks. Core IDM (HIV, HBV, HCV, Syphilis) laboratory results are fed directly into the system, locking reactives at the database layer to quarantine (SOP 3) and automatically defers the donor in MDM.
M2: Inventory & Prep	SOP 2, 4	Manages centrifugation and splitting parents into child components (PRBC, FFP, PLT). Ensures parent-child lineage. Restricts release to AVAILABLE stock by enforcing a physical verify-scan against printed ISBT-128 barcodes to prevent label mismatch (SOP 4).
M3: Ordering & Logistics	SOP 5, 10	Coordinates hospital B2B orders using HICI AI priority scoring. Controls STAT urgent notifications and emergency MTP uncrossmatched Break-glass releases. Tracks transport boxes in real-time via IoT sensors; auto-holds blood bags experiencing temperature excursions (>6°C or <2°C for >30 minutes).
M4: Crossmatch & Bedside	SOP 6, 7, 8	Enforces 72-hour specimen freshness limits. Blocks electronic crossmatching (EXM) when patient has a positive antibody screening history (SOP 7). Tracks issue record timestamps; server-side chronometer overrides nurse check if returned after 30 minutes, marking units as WASTED (SOP 8). Bedside verification (SOP 6) stops transfusions instantly if ABO/Rh mismatches are scanned.
M5: Hemovigilance	SOP 9	Logs post-transfusion reactions. Instantly launches automated bidirectional Lookback: quarantines all other sibling components from the same donation ID (DIN) in stock/transit and suspends donor eligibility in MDM to prevent multiple exposures (SOP 9).

IV. GRANULAR STAGE OPERATIONAL & CLINICAL SAFETY ANALYSIS

01

SOP 1: DONOR QUALIFICATION & INTERVAL CHECKS

OPERATIONAL FLOW

Donor registries scan CCCD identity chips.
Physical checks record hemoglobin, weight, and blood pressure. Core database scans history and deferral blacklists.

DATA I/O & STATE TRANSITIONS

Input: CCCD data, age, weight, donation type.
Output: eligibilityStatus. DB: writes Donors record, generates DIN code, status = COLLECTED.

SAFETY CDSS & GUARDRAILS

Circular 26 Compliance: blocks donors under 18 or over 60, males <45kg, females <42kg.
Enforces whole blood interval of 12 weeks, platelets of 2 weeks.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

TECHNICAL RISK: Donation checks must execute on the backend API. If done solely on the frontend, an API interceptor can bypass interval rules, introducing high-risk donors.

02

SOP 2 & 3: COMPONENT PREP, LIMS & QUARANTINE GATING

OPERATIONAL FLOW

High-speed centrifugation separates Whole Blood into PRBC, FFP, and PLT components. LIMS feeds HIV, HBV, HCV, and Syphilis assay metrics directly.

DATA I/O & STATE TRANSITIONS

Input: parentDin, componentTypes. Output: childUnitIds. DB: creates BloodComponents child tree, status = QUARANTINE.

SAFETY CDSS & GUARDRAILS

Quarantine Gate: restricts any reservations or issuances. Clears to AVAILABLE only if all IDM markers are Non-Reactive. Reactive results automatically trigger immediate DISCARDED and quarantine related co-components.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

CONCURRENCY RACE: Saving LIMS updates and changing status must run inside a single ACID Database Transaction with a row-level lock (SELECT FOR UPDATE) on all co-components.

03

SOP 4: ISBT-128 PRINT & FORCE VERIFY-SCAN

OPERATIONAL FLOW

Once cleared, standard ISBT-128 label prints. Medical staff must scan the physical label barcode and match it against the digital record prior to shelving.

DATA I/O & STATE TRANSITIONS

Input: scannedDin, operatorId. Output: verificationStatus. DB: updates status = AVAILABLE, logs AuditEvent.

SAFETY CDSS & GUARDRAILS

Verify-Scan Defenses: blocks shelf registration if barcodes do not match digital records (e.g. A-type bag labeled as B-type), routing unit to an isolated HOLD state.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

DATA DRIFT: If a manager updates bag attributes after printing but before verify-scan, the barcode mismatch must fail. Verify-Scan API must query live DB state.

04

SOP 10: B2B ORDER, HICI AI & STAT PRIORITIZATION

OPERATIONAL FLOW

Hospitals place orders online. AI HICI rates urgency. Emergency STAT triggers alarms. Clinicians activate Massive Transfusion Protocol (MTP) for trauma.

DATA I/O & STATE TRANSITIONS

Input: Qty, BloodType, priority, isBreakGlass.
Output: orderId, reservedComponents. DB: creates OrderItems, status = RESERVED.

SAFETY CDSS & GUARDRAILS

MTP Break-Glass Bypass: standard MTP blocks uncrossmatched release if specimen has expired (>72h) or patient has antibodies. Setting isBreakGlass = true bypasses checks, forcing O-type release, but logs un-reconciled cases for retrospective review.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

OVER-ALLOCATION RISK: Multiple operators dispatching blood can cause double-reservations. The backend must enforce Optimistic Locking (version column check) on inventory tables.

05

SOP 5: LOGISTICS & IOT TELEMETRY TEMPERATURE TRACKING

OPERATIONAL FLOW

Blood bags are placed in transport boxes equipped with continuous IoT temperature loggers. Couriers scan QR driver tokens to dispatch.

DATA I/O & STATE TRANSITIONS

Input: courierToken, scannedDin, temperature readings. Output: telemetryStatus. DB: writes TelemetryLogs, status = IN_TRANSIT.

SAFETY CDSS & GUARDRAILS

Excursion Lock: if IoT box temperature drifts outside 2-6°C for a cumulative >30 minutes, an excursion alarm fires, locking the bags as WASTED upon hospital arrival.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

OFFLINE BLINDSPOT: If cellular signal drops, temperature drifts are missed. System must require receiving terminal to download local data logger logs to verify.

06

SOP 7: RECIPIENT SPECIMEN REGISTRY & COMPATIBILITY MATCHER

OPERATIONAL FLOW

Hospital lab receives blood bags and registers patient samples. Runs Immediate Spin (IS), Antiglobulin (AHG), or Electronic (EXM) crossmatching.

DATA I/O & STATE TRANSITIONS

Input: componentId, patientId, method, specimenDate. Output: crossmatchResult. DB: creates Crossmatches, status = CROSSMATCHED.

SAFETY CDSS & GUARDRAILS

AABB SOP 7 Gating: validates specimen date; blocks if >72h. If the patient has a positive antibody screen history, EXM and IS are strictly blocked, forcing AHG manual crossmatching.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

OUT-OF-SYNC ANTIBODIES: If patient antibody records in LIS have a sync lag, EXM might pass on a sensitized patient. Crossmatch API must invoke live LIS queries.

07

SOP 8: HOSPITAL ISSUANCE & 30-MINUTE GOLDEN RULE

OPERATIONAL FLOW

Crossmatched bags are issued to ward nurses. If the transfusion is delayed, the bag must be returned to the blood bank within 30 minutes.

DATA I/O & STATE TRANSITIONS

Input: issueRecordId, coldChainOk, visualOk. Output: returnStatus. DB: updates IssueRecord, status = AVAILABLE / WASTED.

SAFETY CDSS & GUARDRAILS

Server Chronometer Override: server computes issue-to-return delta. If >30 minutes, it overrides manual approval, forces Timeout status, and wastes the unit.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

CLOCK DRIFT: If local hospital servers have clock desynchronization, the 30-min window is compromised. Servers must enforce NTP sync and compute using UTC.

08

SOP 6: BEDSIDE DOUBLE-VERIFICATION & TRANSFUSION

OPERATIONAL FLOW

At the bedside, two qualified nurses scan the patient's wristband, the blood bag barcode, and their nurse ID badges before transfusion.

DATA I/O & STATE TRANSITIONS

Input: patientMrn, scannedDin, nurseld1, nurseld2. Output: matcherStatus. DB: creates Transfusions, status = TRANSFUSED.

SAFETY CDSS & GUARDRAILS

Rigid Bedside Matcher: blocks transfusion if ABO/Rh is incompatible, if the bag is not RESERVED for this patient, or if status is not ISSUED. No bypass button allowed.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

OFFLINE LATENCY: If nurse PDA is offline, local cache is used. If a bag was recalled centrally while offline, the scan might pass. Local cache must have a strict <2h TTL.

09

SOP 9: ADVERSE REACTION REPORT & BIDIRECTIONAL LOOKBACK

OPERATIONAL FLOW

If the patient shows fever, hemolytic reaction, or hives during transfusion, the nurse stops the flow and reports a Hemovigilance reaction.

DATA I/O & STATE TRANSITIONS

Input: transfusionId, symptoms, severity. Output: lookbackTriggered. DB: writes AdverseReactions, status = TRANSFUSED (with reaction flag).

SAFETY CDSS & GUARDRAILS

Automated Bidirectional Lookback: immediately quarantines all sibling components (FFP, PLT from same DIN) in stock or transit, and temporarily defers the donor in MDM.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

ASYNC BREAKAGE: If Lookback runs as a background task and fails (OOM or db timeout), sibling bags remain active. Lookback must run inside the main transaction block.

OPERATIONAL FLOW

During network drops, edge nodes log scans locally. Upon reconnection, Sync Manager pushes events chronologically to central servers.

DATA I/O & STATE TRANSITIONS

Input: offlineEventsList. Output: syncStatus. DB: reconciles SyncQueue, updates Inventory and Issues.

SAFETY CDSS & GUARDRAILS

Suspended Conflict Routing: if a bag is issued offline but centrally transferred, the central server blocks Last-Write-Wins, marks status = SUSPENDED_CONFLICT, and alerts QA.

TECHNICAL VULNERABILITY & CONCURRENCY DIAGNOSTICS

DATA CORRUPTION: Network latency can replay events out of order. System must sort events by strict client physical scanned_at timestamp to rebuild lineage.